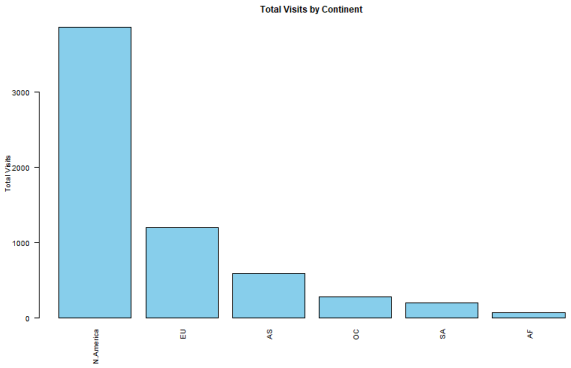
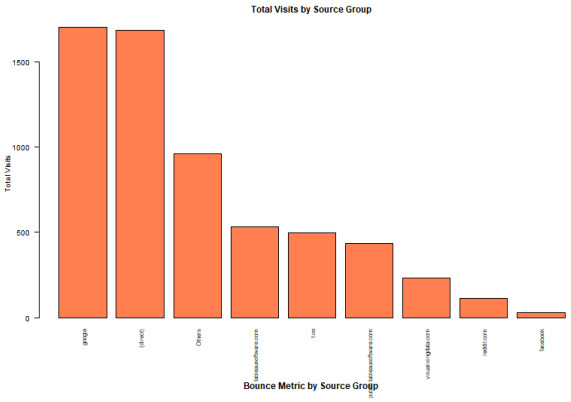
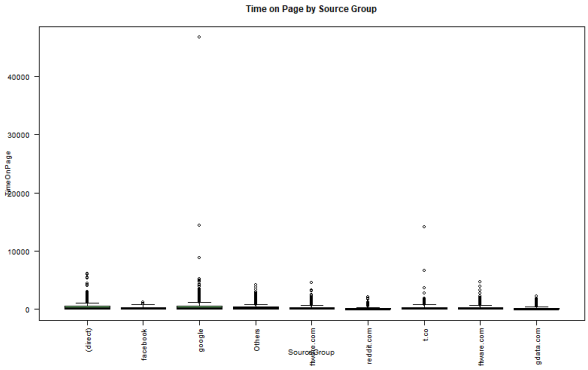
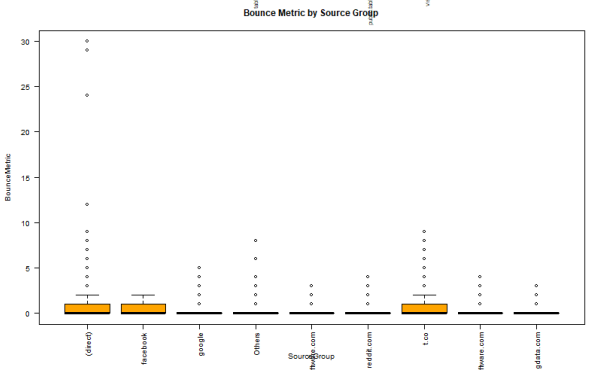
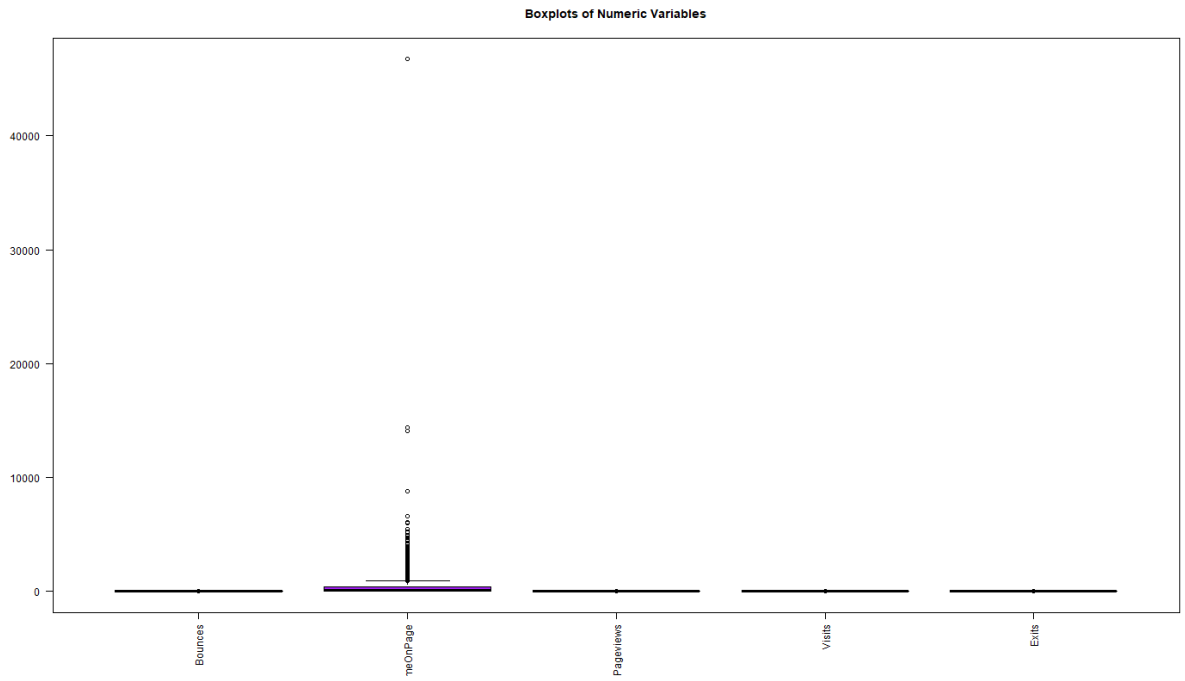
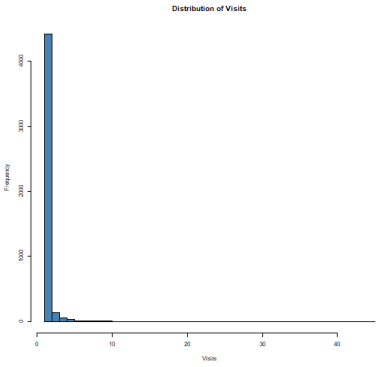

Total Visits by Continent
Total Visits
N America
EU
AS
OC
SA
AF

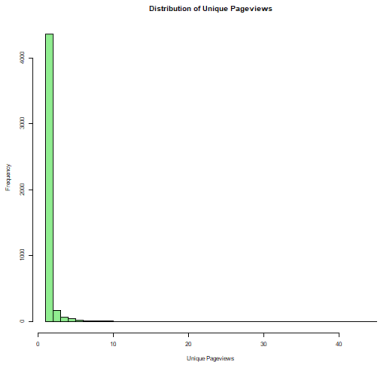

Total Visits by Source Group
Total Visits
google
(direct)
Others
t.co

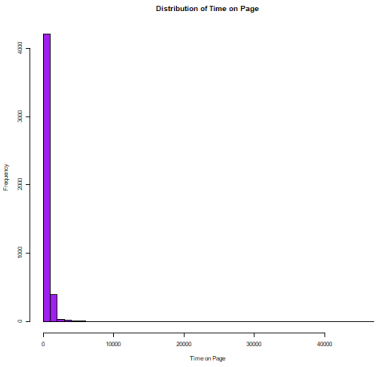

Time on Page by Source Group
TimeOnPage
SourceGroup
(direct)
facebook
google
Others
t.co

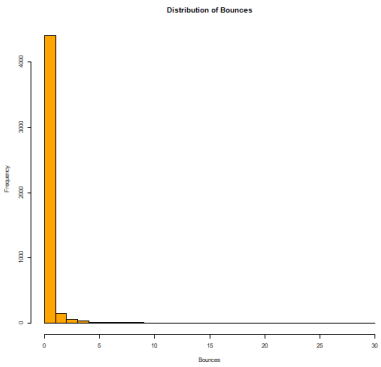

Bounce Metric by Source Group
BounceMetric
SourceGroup
(direct)
facebook
google
Others
t.co

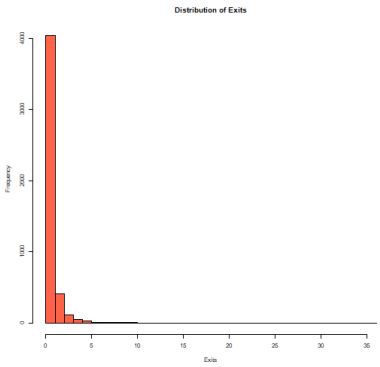

Boxplots of Numeric Variables
Bounces
TimeOnPage
Pageviews
Visits
Exits





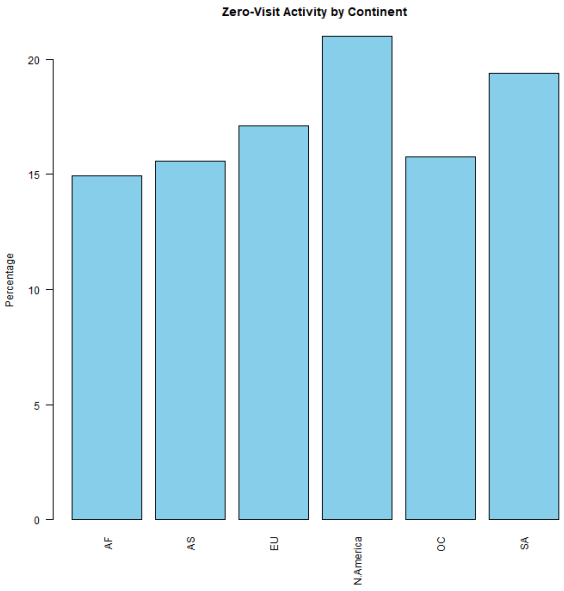


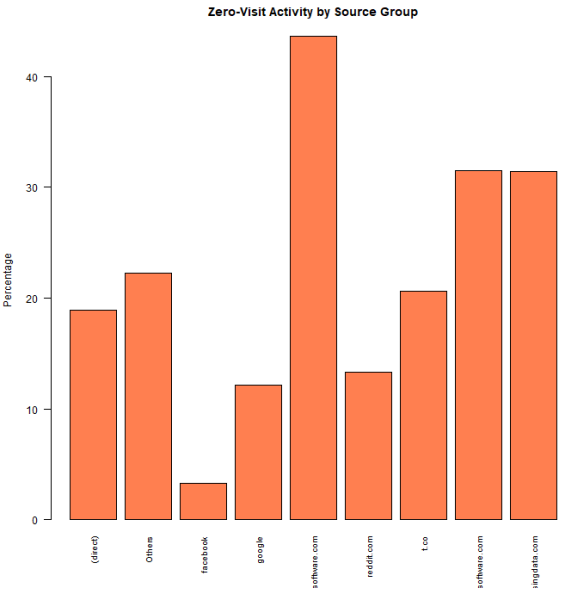


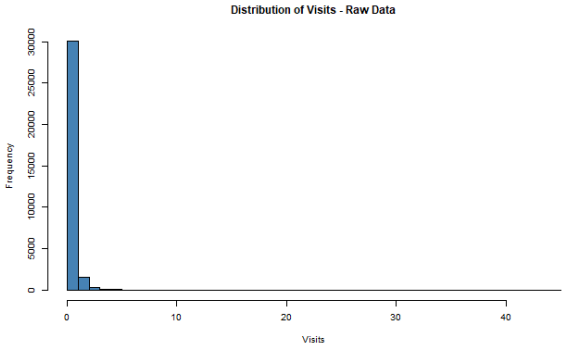


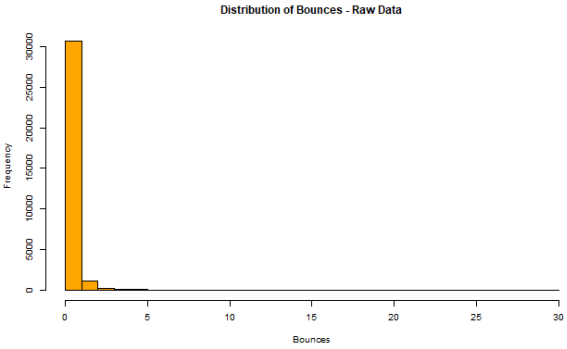
Task 1 Summary

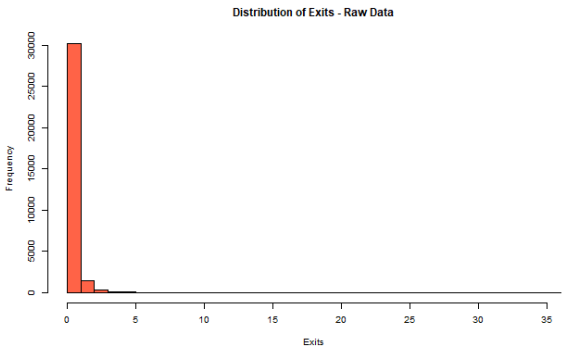
Records: 4679
Total Visits: 6193
Mean Visits: 1.32
Mean Time on Page: 358.31
Mean Bounces: 0.31
Mean Exits: 0.85





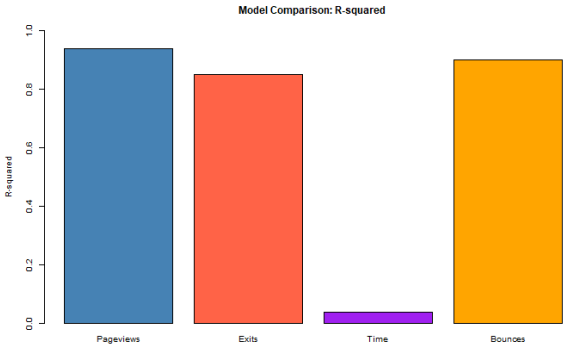


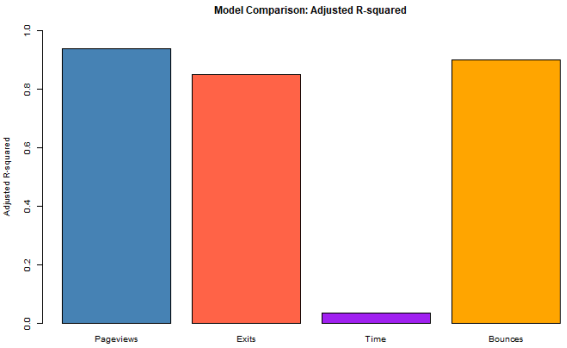




Data Quality Summary

Rows: 32109
Columns: 8
Duplicate rows: 25138
Rows with Visits = 0: 6223
Missing values found: 0





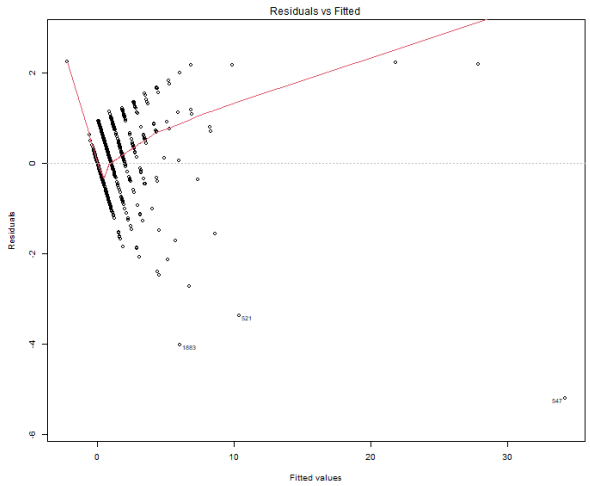
	BusinessQTask
1	Task 2: Unique Pageviews depend on Visits
2	Task 3: Factors affecting Exits
3	Task 4: Factors affecting Time on Page
4	Task 5: Factors impacting Bounces

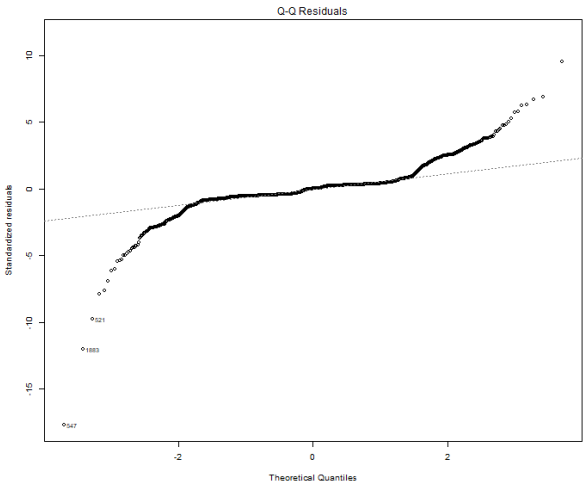
	Model
1	UniquePageviews ~ Visits
2	Exits ~ Bounces + TimeOnPage + UniquePageviews + Visits + Continent + SourceGroup
3	TimeOnPage ~ Bounces + Exits + UniquePageviews + Visits + Continent + SourceGroup
4	BounceMetric ~ TimeOnPage + Exits + UniquePageviews + Visits + Continent + SourceGroup

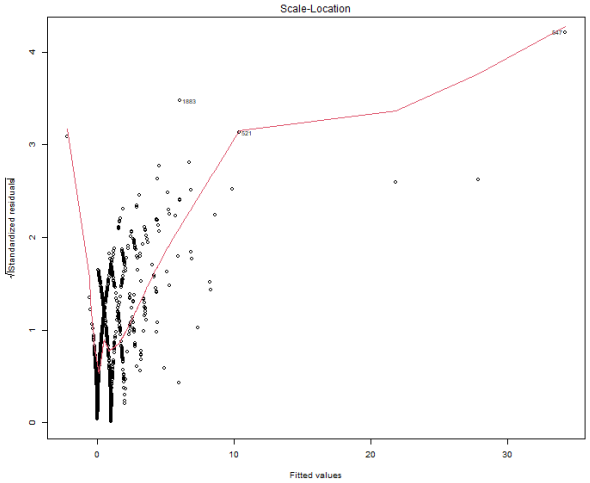
	R_Squared	Adjusted_R_Squared	P_Value
1	0.936197	0.936165	0.000000e+00
2	0.847879	0.847823	0.000000e+00
3	0.336302	0.335924	3.256941e-28
4	0.897506	0.897176	0.000000e+00

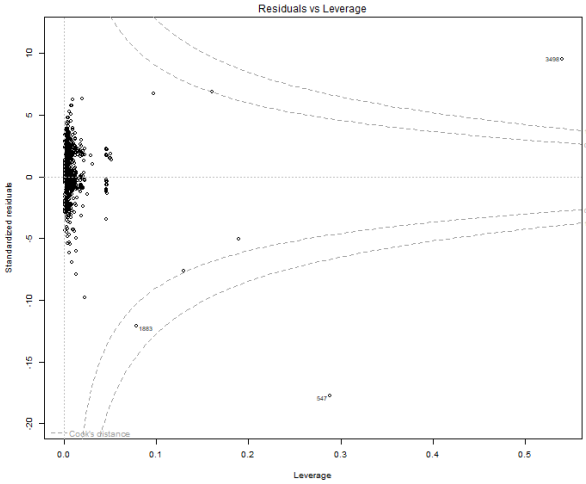
Business Questions Answered

1. Variables summarised using descriptive statistics and plots.
2. Unique pageviews tested against visits using correlation and regression.
3. Exits modelled using engagement, traffic, continent, and source variables.
4. Time on page modelled using traffic and engagement variables.
5. Bounces modelled using traffic, exits, time, continent, and source group.


Residuals vs Fitted
Residuals
Fitted values
1883
521
547


Q-Q Residuals
Standardized residuals
Theoretical Quantiles
1883
521
547


Scale-Location
√|Standardized residuals|
Fitted values
1883
521
547


Residuals vs Leverage
Standardized residuals
Leverage
Cook's distance
1883
547

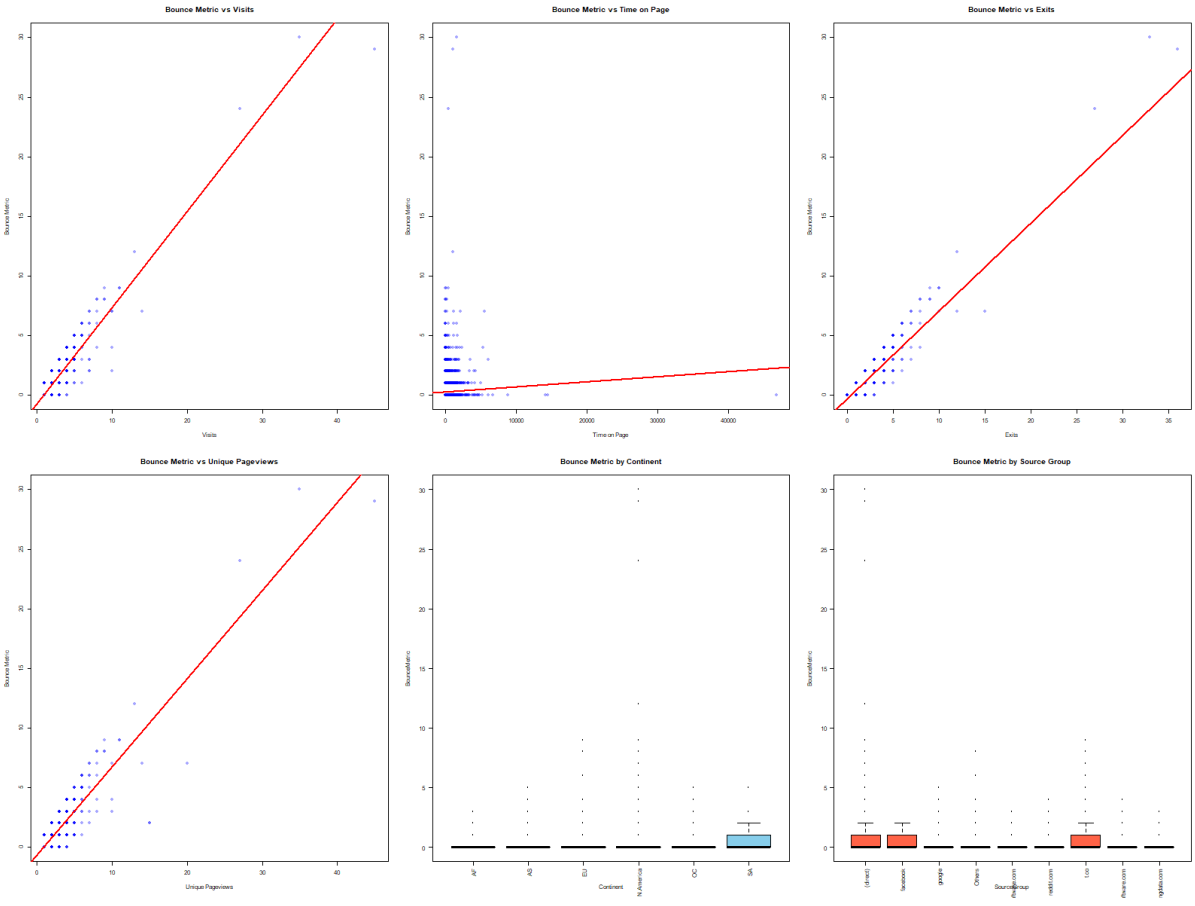
```
Call:
lm(formula = BounceMetric ~ TimeOnPage + Exits + UniquePageviews +
    Visits + Continent + SourceGroup, data = clean_data)

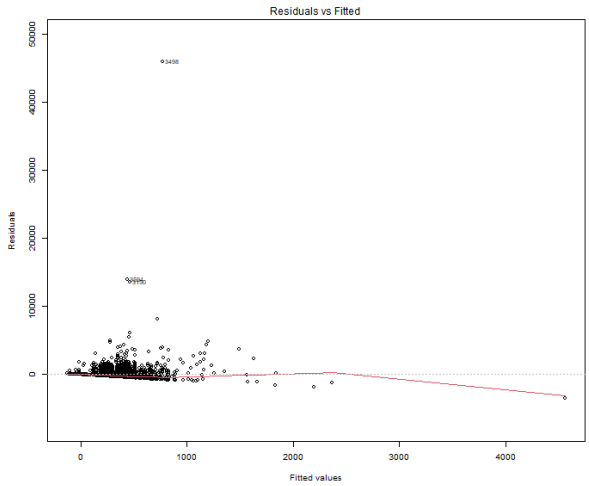
Residuals:
    Min      1Q  Median      3Q     Max
-5.2853 -0.1629  0.0888  0.1139  2.2488

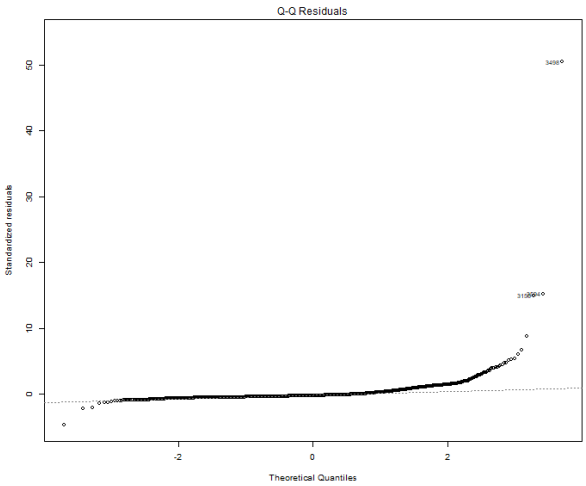
Coefficients:

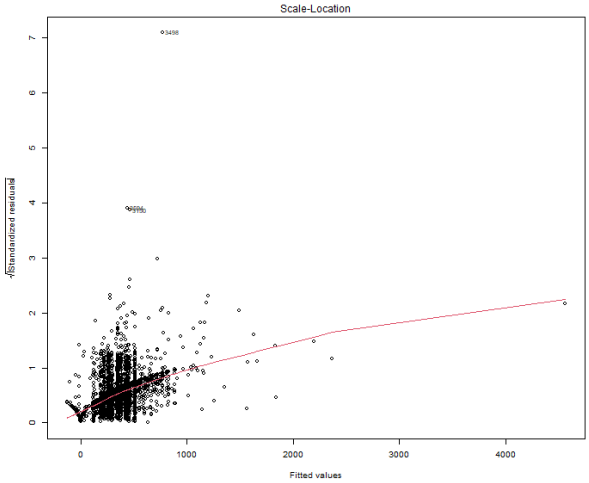
                                        Estimate Std. Error t value Pr(>|t|)
(Intercept)                           -5.304e-01  4.747e-02 -11.173  < 2e-16 ***
TimeOnPage                            -5.432e-05  5.508e-06  -9.862  < 2e-16 ***
Exits                                  2.774e-01  9.012e-03  30.776  < 2e-16 ***
UniquePageviews                       -4.644e-02  1.582e-02  -2.935  0.00335 **
Visits                                 6.085e-01  1.662e-02  36.129  < 2e-16 ***
ContinentAS                           -1.066e-01  4.866e-02  -2.191  0.02851 *
ContinentEU                           -1.130e-01  4.739e-02  -2.385  0.01714 *
ContinentN.America                    -1.251e-01  4.668e-02  -2.684  0.00730 **
ContinentOC                           -2.744e-02  5.141e-02  -0.534  0.59356
ContinentSA                            1.709e-02  5.423e-02   0.315  0.75267
SourceGroupfacebook                    2.228e-01  7.502e-02   2.970  0.00299 **
SourceGroupgoogle                     -2.679e-02  1.444e-02  -1.856  0.06356 .
SourceGroupothers                      8.191e-03  1.645e-02   0.498  0.61856
SourceGrouppublic.tableausoftware.com -9.432e-03  2.197e-02  -0.429  0.66773
SourceGroupreddit.com                  1.170e-01  3.848e-02   3.041  0.00237 **
SourceGroupt.co                        8.796e-02  2.213e-02   3.974 7.16e-05 ***
SourceGrouptableausoftware.com         2.522e-02  1.986e-02   1.270  0.20421
SourceGroupvisualisingdata.com         8.249e-02  2.744e-02   3.006  0.00266 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

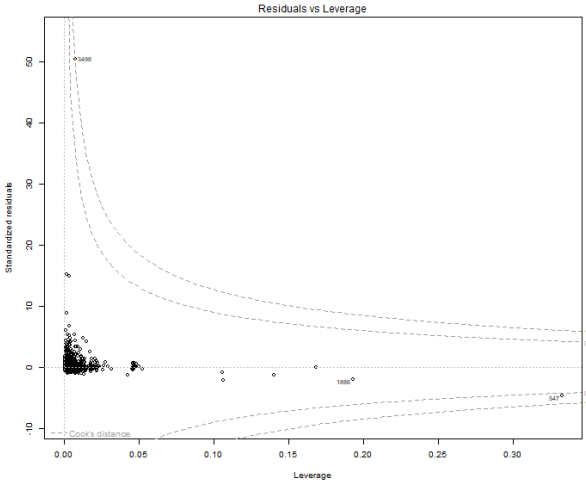
Residual standard error: 0.3477 on 4661 degrees of freedom
Multiple R-squared:  0.8975,Adjusted R-squared:  0.8971
F-statistic:  2401 on 17 and 4661 DF,  p-value: < 2.2e-16
```




Residuals vs Fitted
Residuals
Fitted values
3498


Q-Q Residuals
Standardized residuals
Theoretical Quantiles
3498


Scale-Location
√|Standardized residuals|
Fitted values
3498


Residuals vs Leverage
Standardized residuals
Leverage
Cook's distance
3498

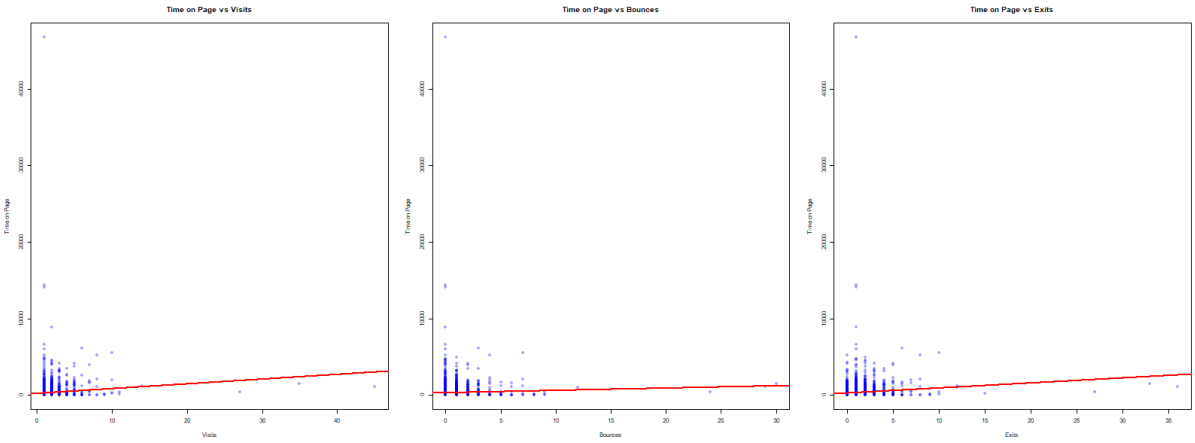
```
Call:
lm(formula = TimeOnPage ~ Bounces + Exits + UniquePageviews +
    Visits + Continent + SourceGroup, data = clean_data)

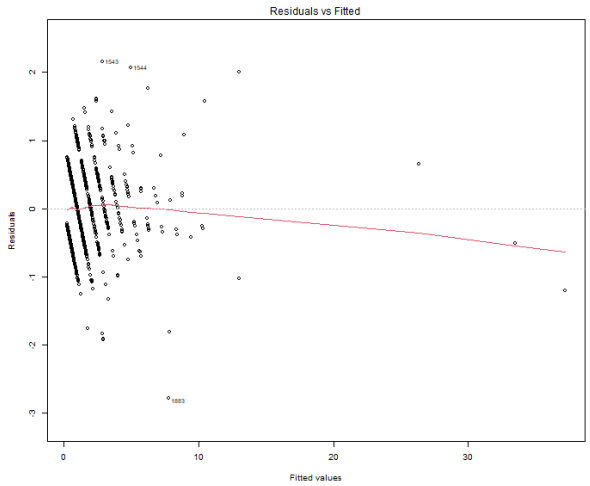
Residuals:
   Min     1Q Median     3Q    Max
 -3585   -298   -166     54  45977

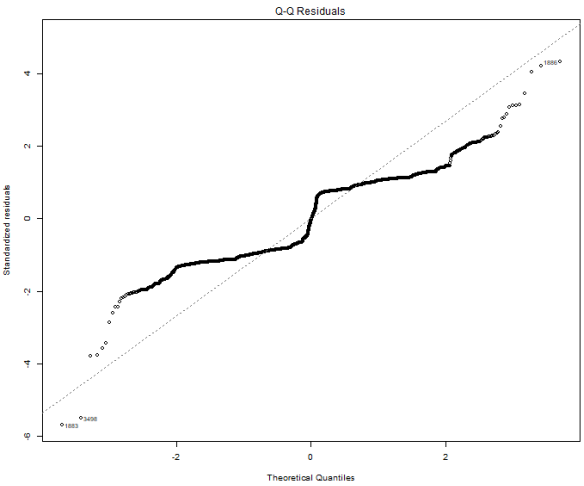
Coefficients:

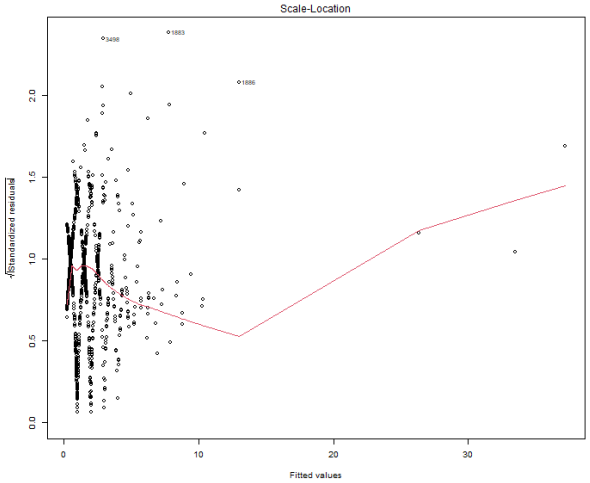
                                          Estimate Std. Error t value Pr(>|t|)
(Intercept)                               -28.943    126.584  -0.229   0.8192
Bounces                                  -376.253     38.152  -9.862  < 2e-16 ***
Exits                                     159.817     25.910   6.168 7.49e-10 ***
UniquePageviews                           -15.397     41.673  -0.369   0.7118
Visits                                    229.521     49.377   4.648 3.44e-06 ***
ContinentAS                                33.283    128.134   0.260   0.7951
ContinentEU                                36.633    124.805   0.294   0.7691
ContinentN.America                        112.298    122.730   0.899   0.3689
ContinentOC                                43.860    135.300   0.324   0.7458
ContinentSA                               366.768    142.608   2.572   0.0101 *
SourceGroupfacebook                       -55.323    197.614  -0.280   0.7795
SourceGroupgoogle                          56.082     38.000   1.476   0.1401
SourceGroupothers                         -32.414     43.293  -0.749   0.4541
SourceGrouppublic.tableausoftware.com     -76.301     57.814  -1.320   0.1870
SourceGroupreddit.com                     -93.085    101.365  -0.918   0.3585
SourceGroupt.co                             5.174     58.342   0.089   0.9293
SourceGrouptableausoftware.com            -29.982     52.281  -0.573   0.5663
SourceGroupvisualisingdata.com            -95.394     72.285  -1.320   0.1870
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

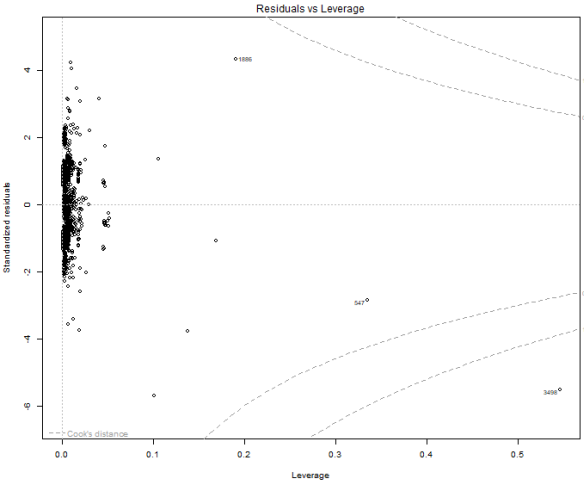
Residual standard error: 915 on 4661 degrees of freedom
Multiple R-squared:  0.03676,Adjusted R-squared:  0.03325
F-statistic: 10.46 on 17 and 4661 DF,  p-value: < 2.2e-16
```




Residuals vs Fitted
Residuals
Fitted values
1543
1544
1883


Q-Q Residuals
Standardized residuals
Theoretical Quantiles
1886
3498
1883


Scale-Location
√|Standardized residuals|
Fitted values
3498
1883
1886


Residuals vs Leverage
Standardized residuals
Leverage
Cook's distance
1886
547
3498

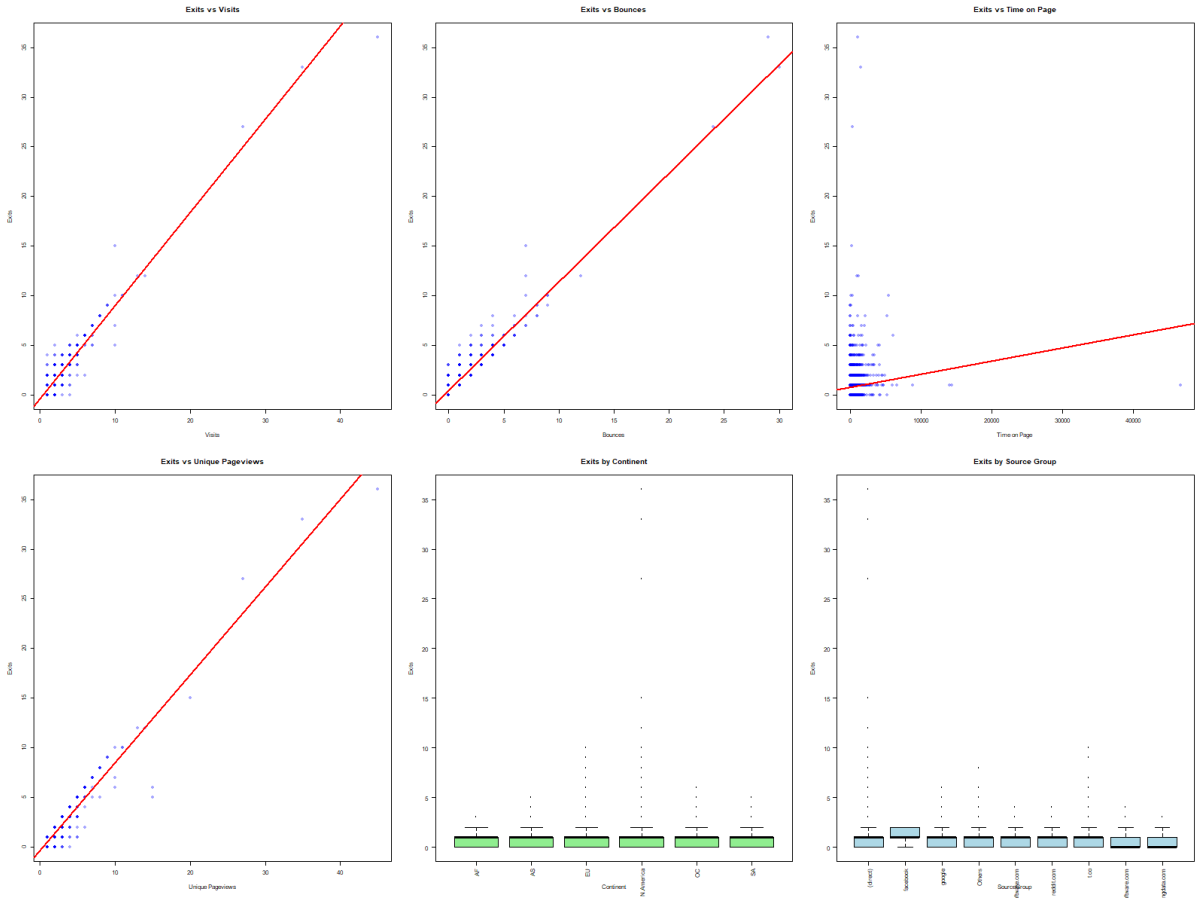
```
Call:
lm(formula = Exits ~ Bounces + TimeOnPage + UniquePageviews +
    Visits + Continent + SourceGroup, data = clean_data)

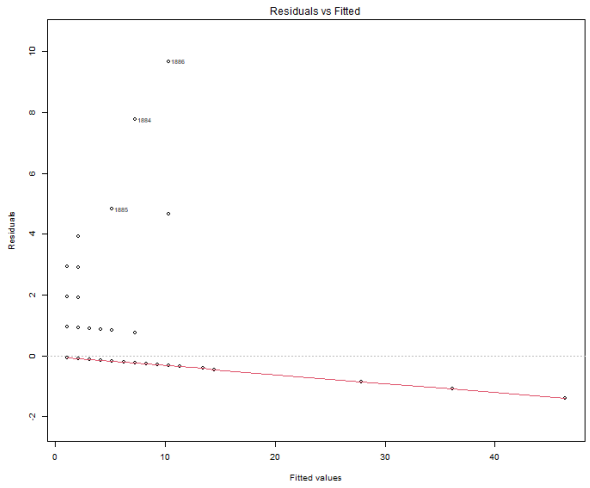
Residuals:
     Min       1Q   Median       3Q      Max
-2.78131 -0.46121 -0.02136  0.47038  2.15825

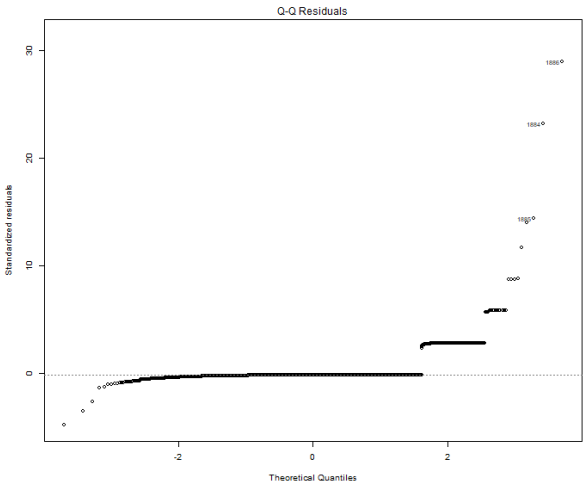
Coefficients:

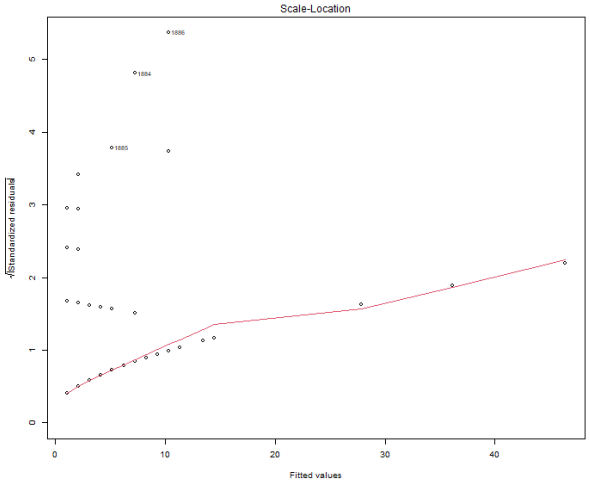
                                          Estimate Std. Error t value Pr(>|t|)
(Intercept)                              9.807e-02  7.126e-02   1.376  0.16878
Bounces                                  6.089e-01  1.979e-02  30.776  < 2e-16 ***
TimeOnPage                               5.066e-05  8.213e-06   6.168 7.48e-10 ***
UniquePageviews                          4.352e-01  2.258e-02  19.271  < 2e-16 ***
Visits                                  -3.338e-03  2.786e-02  -0.120  0.90466
ContinentAS                             -1.471e-02  7.214e-02  -0.204  0.83845
ContinentEU                             -1.992e-02  7.027e-02  -0.283  0.77686
ContinentN.America                      -4.477e-02  6.919e-02  -0.648  0.51716
ContinentOC                             -4.098e-02  7.618e-02  -0.538  0.59064
ContinentSA                             -7.929e-02  8.034e-02  -0.987  0.32371
SourceGroupfacebook                      1.532e-01  1.112e-01   1.377  0.16863
SourceGroupgoogle                        9.837e-02  2.136e-02   4.231 2.37e-05 ***
SourceGroupothers                       -7.176e-02  2.435e-02  -2.946  0.00323 **
SourceGrouppublic.tableausoftware.com   -5.529e-02  3.255e-02  -1.696  0.08993 .
SourceGroupreddit.com                   -1.410e-01  5.704e-02  -2.472  0.01349 *
SourceGroupt.co                         -7.825e-02  3.283e-02  -2.140  0.03243 *
SourceGrouptableausoftware.com          -1.524e-01  2.935e-02  -5.192 2.17e-07 ***
SourceGroupvisualisingdata.com          -2.384e-01  4.856e-02  -5.879 4.42e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

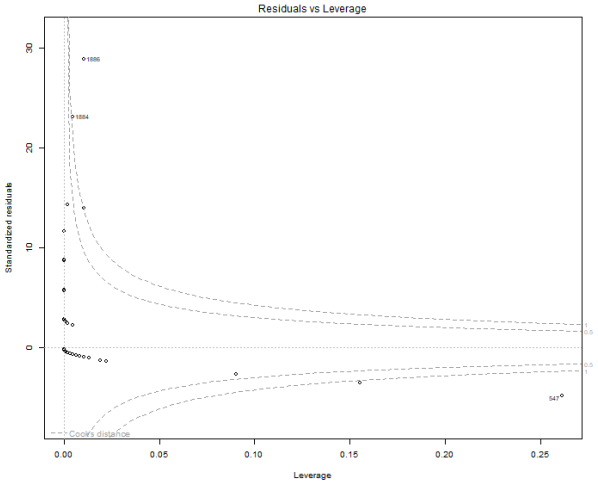
Residual standard error: 0.5152 on 4661 degrees of freedom
Multiple R-squared:  0.8479,Adjusted R-squared:  0.8473
F-statistic:  1528 on 17 and 4661 DF,  p-value: < 2.2e-16
```









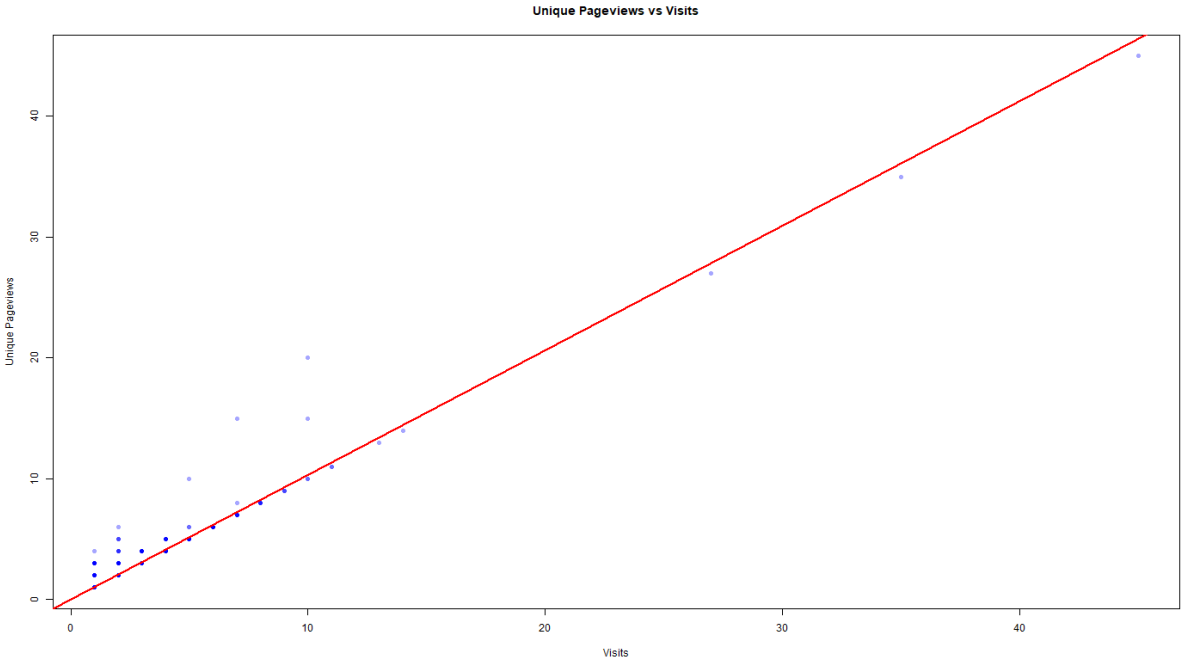


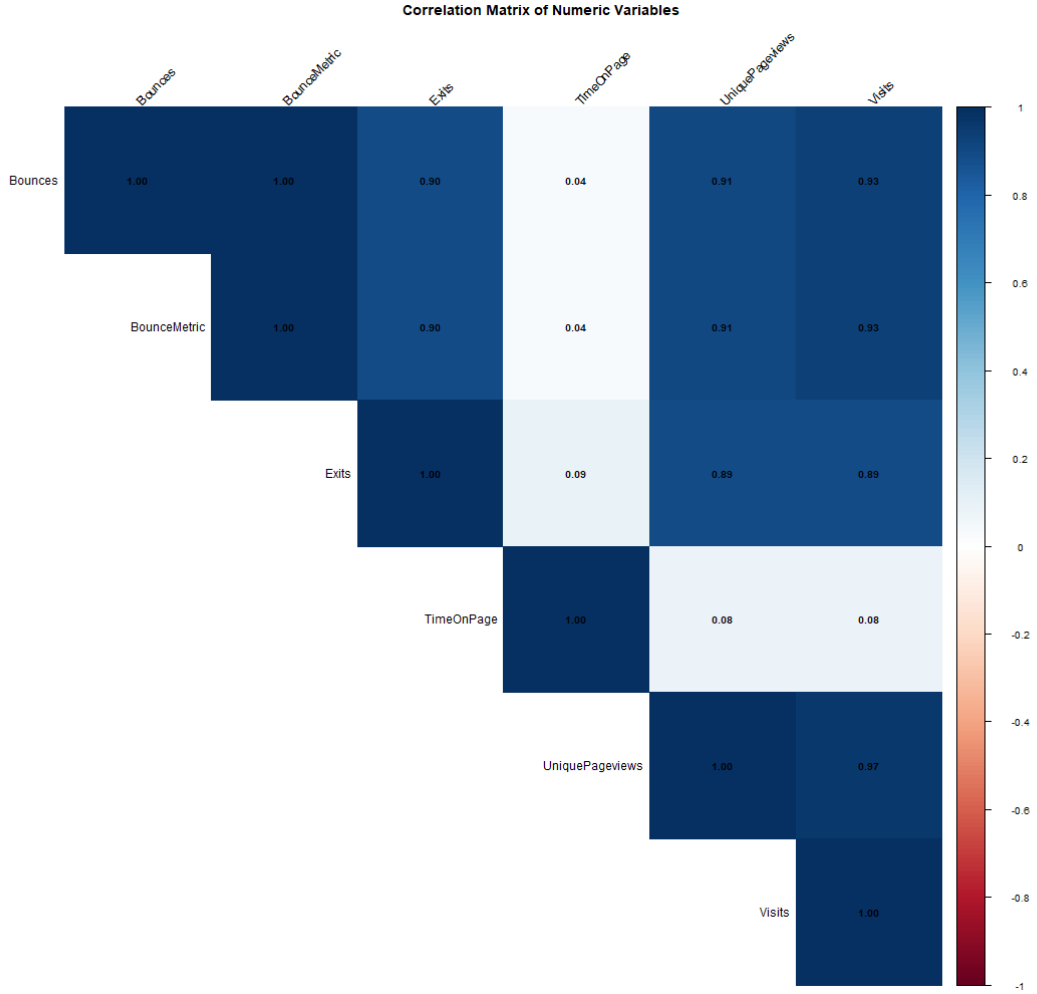

```
Call:
lm(formula = UniquePageviews ~ Visits, data = clean_data)

Residuals:
    Min      1Q  Median      3Q     Max
-1.4018 -0.0544 -0.0544 -0.0544  9.6700

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.023800   0.007162   3.323 0.000897 ***
Visits      1.030622   0.003935 261.918  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3262 on 4677 degrees of freedom
Multiple R-squared:  0.9362,Adjusted R-squared:  0.9362
F-statistic: 6.86e+04 on 1 and 4677 DF,  p-value: < 2.2e-16
```


Unique Pageviews vs Visits
Unique Pageviews
Visits
0
10
20
30
40


Correlation Matrix of Numeric Variables
Bounces
BounceMetric
Exits
TimeOnPage
UniquePageviews
Visits
1.00 1.00 0.90 0.04 0.91 0.93
1.00 0.90 0.04 0.91 0.93
1.00 0.09 0.89 0.89
1.00 0.08 0.08
1.00 0.97
1.00
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1